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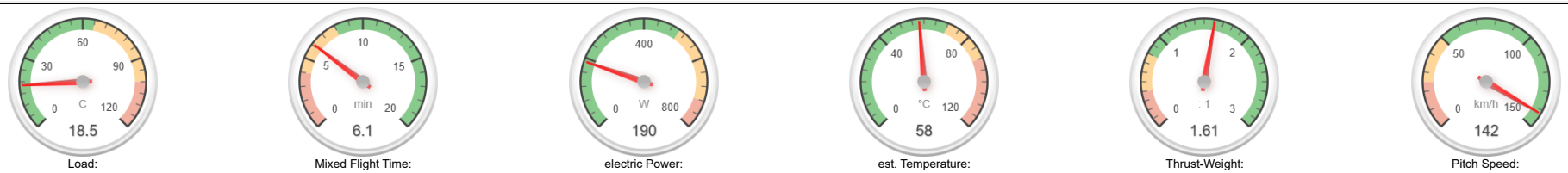
General	Model Weight: 520 g incl. Drive; 18.3 oz	# of Motors: 1 (on same Battery)	Wingspan: 762 mm; 30.00 inch	Wing Area: 17.42 dm²; 270 in²	Drag: coeffcient; 0.18 Cd	Field Elevation: 0 m.ASL; 0 ft.ASL	Air Temperature: 25 °C; 77 °F	Pressure (QNH): 1013 hPa; 29.91 inHg
Battery Cell	Type (Cont. / max. C) - charge state: custom - normal	Configuration: 3 S 1 P	Cell Capacity: 1000 mAh; 1000 mAh total	max. discharge: 85%	Resistance: 0.0113 Ohm	Voltage: 3.7 V	C-Rate: 65 C cont.; 100 C max	Weight: 34 g; 1.2 oz
Controller	Type - Timing: max 20A - normal	Current: 20 A cont.; 20 A max	Resistance: 0.01 Ohm	Weight: 25 g; 0.9 oz	Battery extension Wire: AWG10=5.27mm²	Length: 152.4 mm; 6 inch	Motor extension Wire: AWG12=3.30mm²	Length: 76.2 mm; 3 inch
Motor	Manufacturer - Type (Kv) - Cooling: (² = discontinued) EMAX - EcoII 2207-2400 (2400); medium; search...	KV (w/o torque): 2400 rpm/V; Prop-Kv-Wizard	no-load Current: 2.2 A @ 10 V	Limit (up to 15s): 720 W	Resistance: 0.041 Ohm	Case Length: 20 mm; 0.79 inch	# mag. Poles: 14	Weight: 34 g; 1.2 oz
Propeller	Type - yoke twist: APC Electric E - 0°	Diameter: 5.25 inch; 133.4 mm	Pitch: 4.25 inch; 108 mm	# Blades: 2	Simulation based on: PConst / TConst; 1.08 / 1.0	Gear Ratio: 1 : 1	Flight Speed: 0 km/h; 0 mph	calculate



Remarks:

- The max. current is just below the continuous limit of the controller. In partial load operation, the controller may overheat.
  We suggest excellent ESC cooling or at least 20% headroom for heavy partial load operation (e.g. 3D).
- Excessive wiring extension between battery and ESC leads to high voltage spikes and may harm your ESC input.
  Recommendation: do the ESC supply without wiring extension or install additional low ESR capacitors to the wiring to protect the ESC input.
- The airflow at the propeller blade will stall. Therefore the static thrust and max. current may not be reached. On ground you will measure *Stall Thrust* as maximum.
- 25.0km/h / 15.5mph - above this airspeed stall at the propeller blade will have disappeared completely.

Battery		**Motor @ Optimum Efficiency**		**Motor @ Maximum**		**Propeller**		**Total Drive**		**Airplane**	
Load:	18.51 C	Current:	20.63 A	Current:	18.51 A	Static Thrust:	838 g	Drive Weight:	177 g	All-up Weight:	520 g
Voltage:	10.47 V	Voltage:	10.16 V	Voltage:	10.25 V		29.6 oz		6.2 oz		18.3 oz
Rated Voltage:	11.10 V	Revolutions*:	21381 rpm	Revolutions*:	21839 rpm	Revolutions*:	21839 rpm	Power-Weight:	395 W/kg	Wing Load:	29.9 g/dm²
Energy:	11.1 Wh	electric Power:	209.5 W	electric Power:	189.8 W	Stall Thrust:	601 g		179 W/lb		9.8 oz/ft²
Total Capacity:	1000 mAh	mech. Power:	170.7 W	mech. Power:	154.1 W		21.2 oz	Thrust-Weight:	1.61 : 1	Cubic Wing Load:	7.2
Used Capacity:	850 mAh	Efficiency:	81.5 %	Efficiency:	81.2 %	avail.Thrust @ 0 km/h:	601 g	Current @ max:	18.51 A	est. Stall Speed:	25 km/h
min. Flight Time:	2.8 min			est. Temperature:	58 °C	avail.Thrust @ 0 mph:	21.2 oz	P(in) @ max:	205.5 W		16 mph
Mixed Flight Time:	6.1 min				136 °F	Pitch Speed:	142 km/h	P(out) @ max:	154.1 W	est. Speed (level):	70 km/h
Weight:	102 g			Rm:	57.2 mΩ		88 mph	Efficiency @ max:	75.0 %		43 mph
	3.6 oz			Controller	152873 epm	Tip Speed:	549 km/h	Torque:	0.07 Nm	est. Speed (vertical):	26 km/h
							341 mph		0.05 lbf.ft		16 mph
		Wattmeter readings		**Thermal Propagation**		specific Thrust:	3.17 g/W	Climb Capacity:	1253 m	est. rate of climb:	7.6 m/s
		Current:	18.51 A	2 5 11 17 s			0.11 oz/W		4111 ft	(~70...75°)	1492 ft/min
		Voltage:	10.47 V	38 46 54 57 °C		n100W:	19491 rpm				
		Power:	193.8 W	101 115 129 134 °F		n10N:	24836 rpm				

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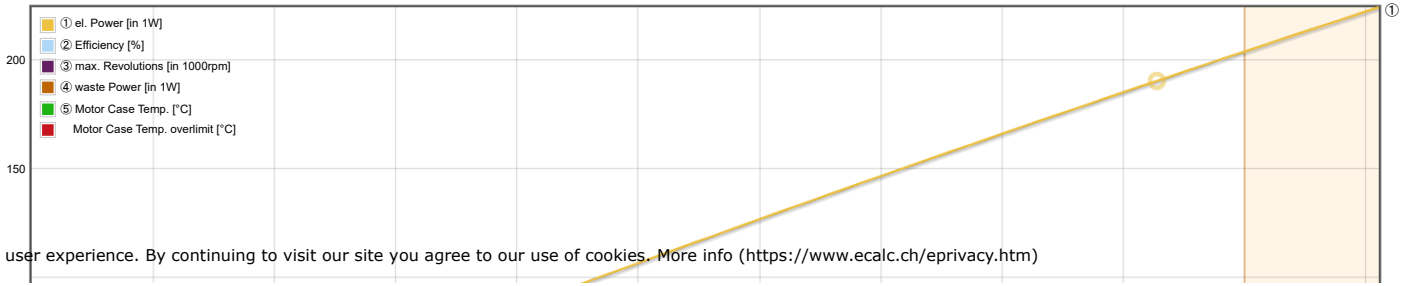
Motor Characteristic at Full Load



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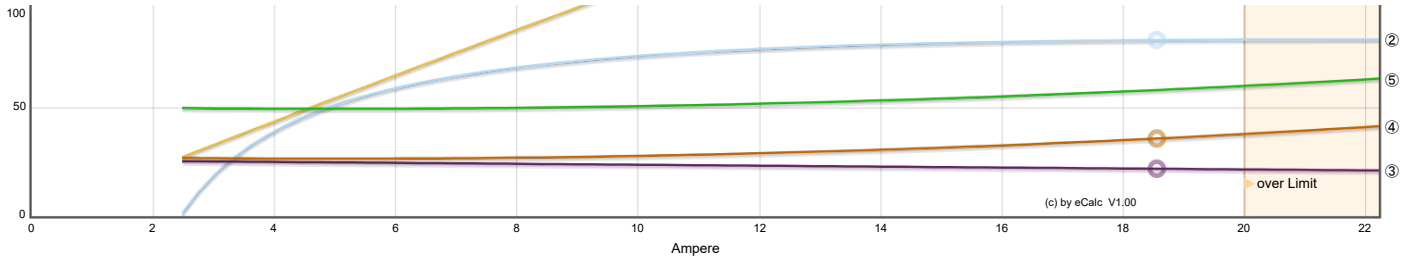
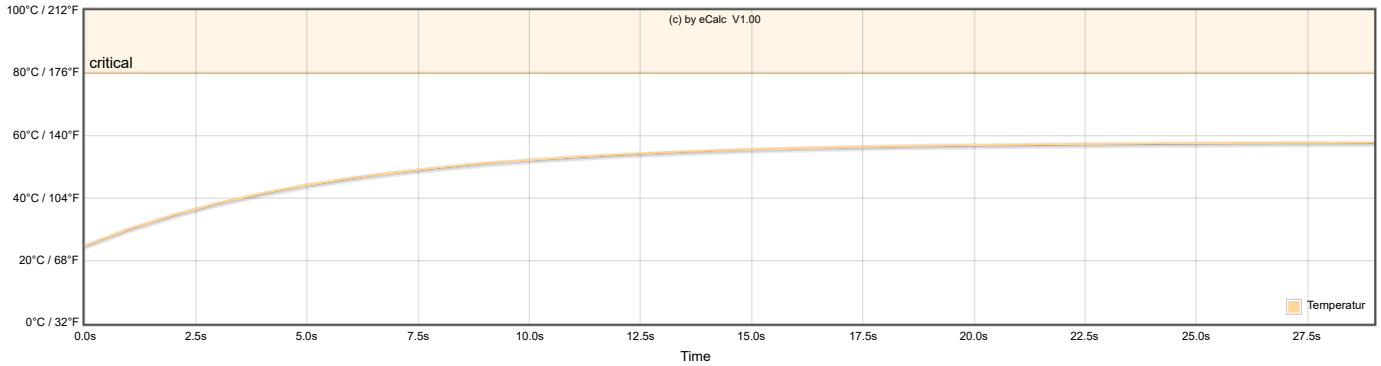


Thermal Propagation at Full Load



Motor Partial Load - Static on Ground

Propeller	Pitch Speed		Throttle	Current (DC)	Voltage (DC)	el. Power	Efficiency	Thrust		Spec. Thrust		Motor Run Time
rpm	km/h	mph	%	A	V	W	%	g	oz	g/W	oz/W	(85%) min
5600	36	23	23	0.5	11.1	5.6	46.6	55	1.9	9.9	0.35	100.2
7000	45	28	29	0.8	11.1	9.0	56.5	86	3.0	9.6	0.34	62.1
8400	54	34	35	1.3	11.1	13.8	63.8	124	4.4	9.0	0.32	40.5
9800	64	39	41	1.9	11.0	20.2	69.0	169	6.0	8.3	0.29	27.5
11200	73	45	47	2.6	11.0	28.6	72.8	220	7.8	7.7	0.27	19.4
12600	82	51	53	3.6	11.0	39.3	75.6	279	9.8	7.1	0.25	14.1
14000	91	56	59	4.9	10.9	52.5	77.6	344	12.1	6.6	0.23	10.5
15400	100	62	66	6.4	10.9	68.6	79.0	417	14.7	6.1	0.21	8.0
16800	109	68	73	8.3	10.8	87.8	80.1	496	17.5	5.6	0.20	6.2
18200	118	73	80	10.5	10.7	110.6	80.9	582	20.5	5.3	0.19	4.9
19600	127	79	87	13.1	10.7	137.2	81.4	675	23.8	4.9	0.17	3.9
21000	**136**	**85**	**95**	**16.2**	**10.5**	**167.9**	**81.8**	**775**	**27.3**	**4.6**	**0.16**	**3.1**
21839	142	88	100	18.5	10.5	189.8	81.2	838	29.5	4.4	0.16	2.8

Motor Partial Load - Dynamic in Flight

Propeller	Speed (level)		Throttle	Current (DC)	Voltage (DC)	el. Power	Efficiency	Thrust		Energy Consumption		Motor Run Time
rpm	km/h	mph	%	A	V	W	%	g	oz	Wh/km	Wh/mi	(85%) min
8048	25	16	32	1.2	11.1	13.0	64.3	91	3.2	0.52	0.83	43.0
9658	30	19	39	1.9	11.0	20.5	70.3	131	4.6	0.68	1.10	27.2
11267	35	22	46	2.8	11.0	30.7	74.4	179	6.3	0.88	1.41	18.0
12877	40	25	53	4.1	11.0	44.2	77.2	234	8.2	1.10	1.78	12.5
14487	45	28	60	5.7	10.9	61.4	79.1	296	10.4	1.36	2.20	8.9
16096	50	31	68	7.8	10.8	82.9	80.4	365	12.9	1.66	2.67	6.6
17706	55	34	76	10.4	10.7	109.2	81.2	442	15.6	1.98	3.20	4.9
19316	60	37	84	13.5	10.6	140.7	81.8	525	18.5	2.35	3.78	3.8
20925	**65**	**40**	**93**	**17.3**	**10.5**	**178.2**	**82.1**	**617**	**21.8**	**2.74**	**4.41**	**2.9**

Important Note:
Before flight recheck your max. current! If your Current, el. Power or RPM are over the manufacturers limits your motor, controller and/or battery may take damage! **Verify before flight by measurment!**

for printing use Landscape format
* The manufacturer limitation is NOT monitored
** Testdata with reduced accuracy

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